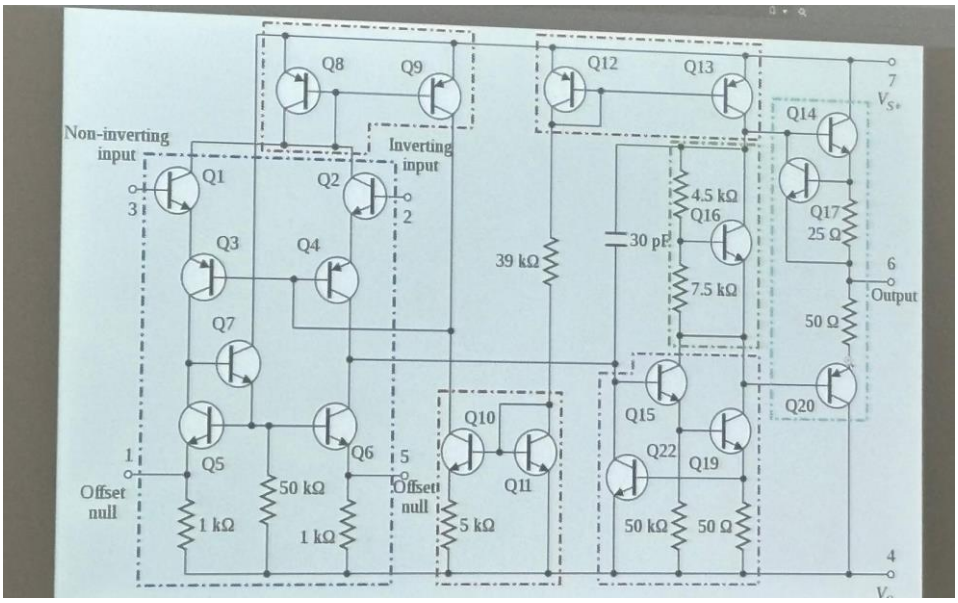
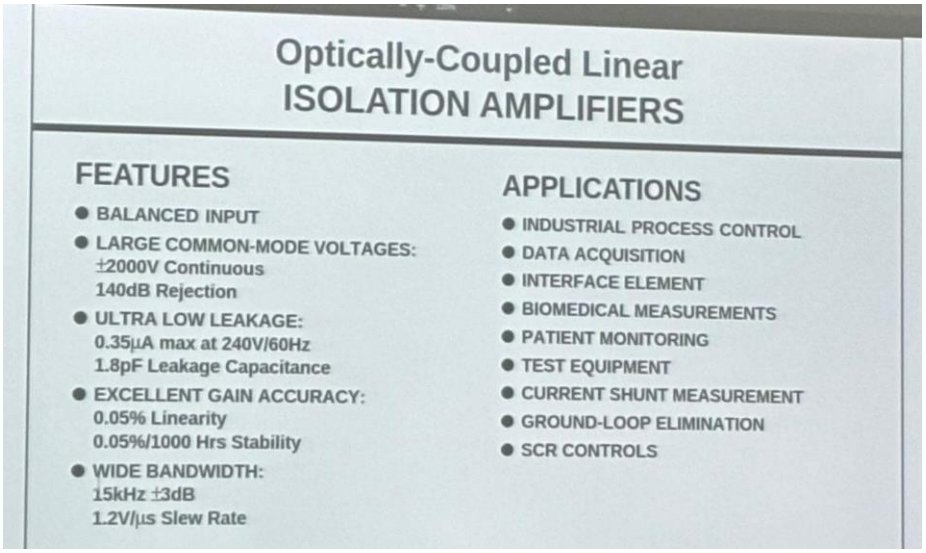
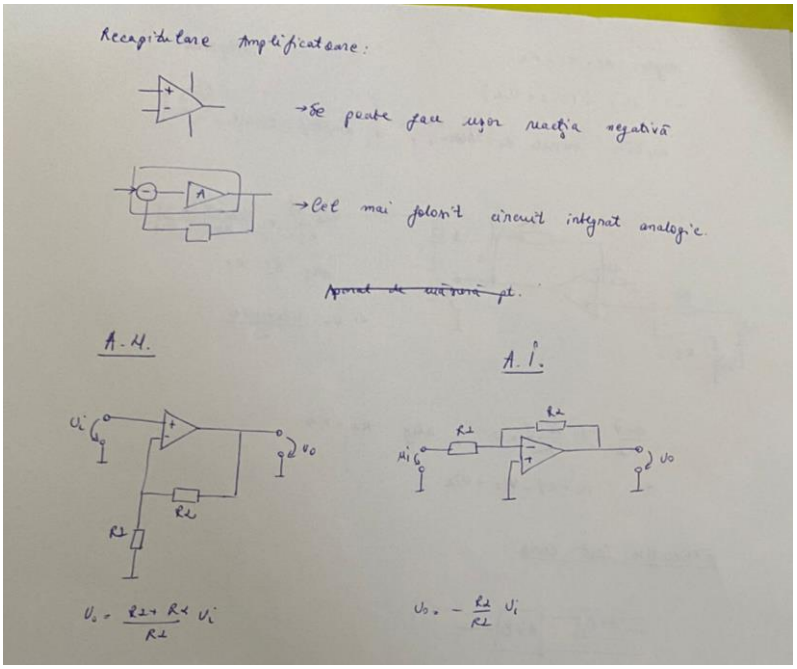
Amplificatorul operational – curs

Recapitluare:



- Reactia negativa este foarte importanta pentru amplificatorul liniar operational
- Cel mai folosit circuit integrat analogic
- Impendanta de intrare foarte mare si impendanta de iesire foarte mica -> circuit ideal
- Q17 – tranzistor important cu rol de protectie
- LA 24 mA amplificatorul se opreste
- Rol de alimentare cu curent constant -> tranzistoarele incercuite de linia punctata





Electrocardiograma :

- Foloseste diferenta de potential pentru a genera graficul ; intrarile aparatului sunt complet izolate de iesiri
- Aparat de masura pentru tensiune

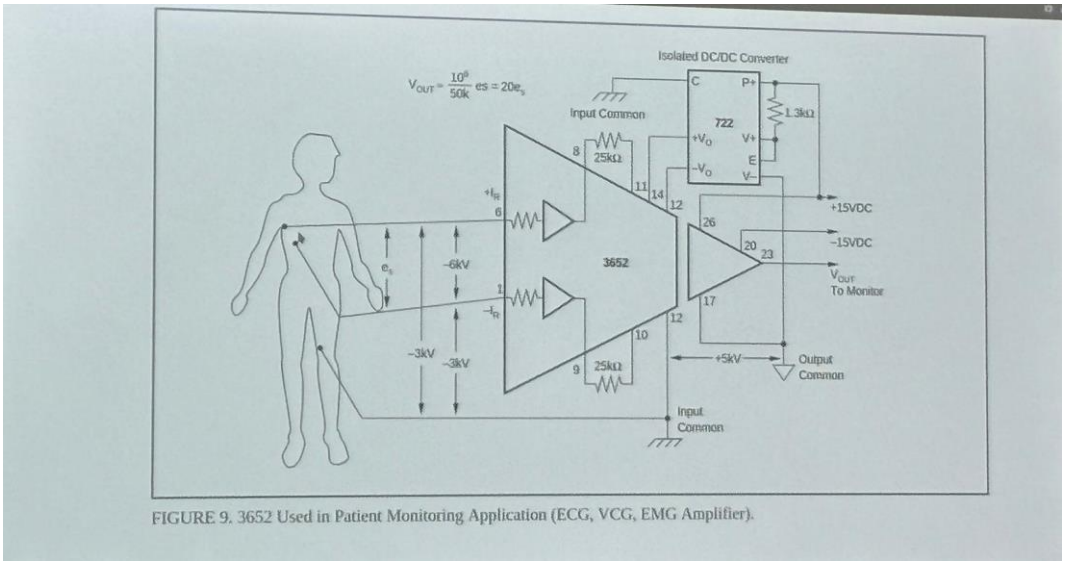
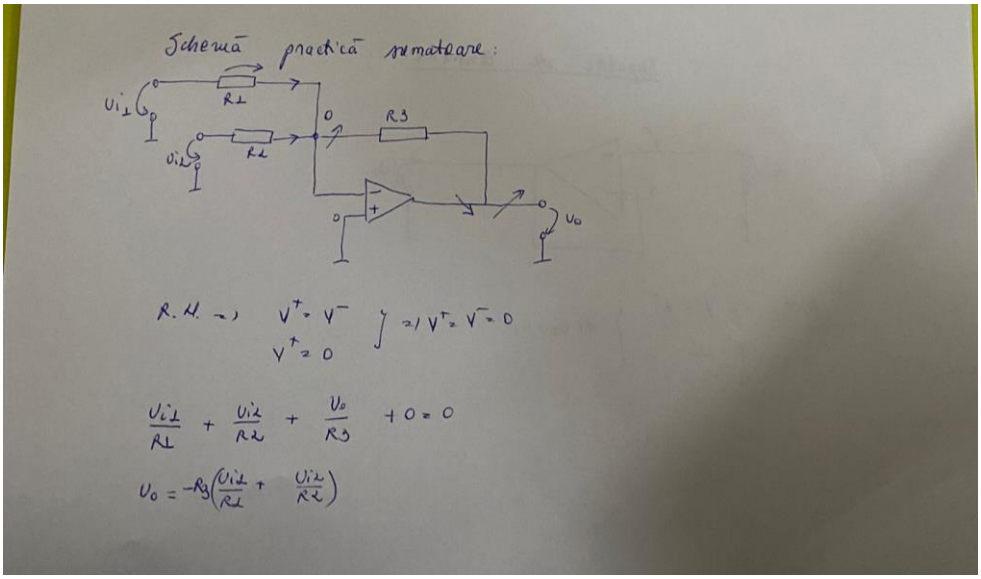
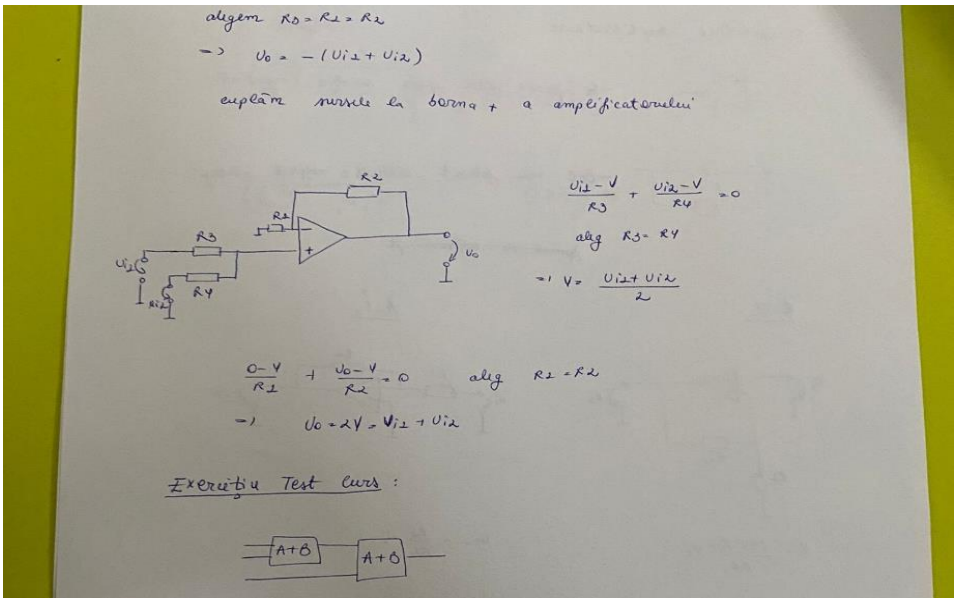


FIGURE 9. 3652 Used in Patient Monitoring Application (ECG, VCG, EMG Amplifier).



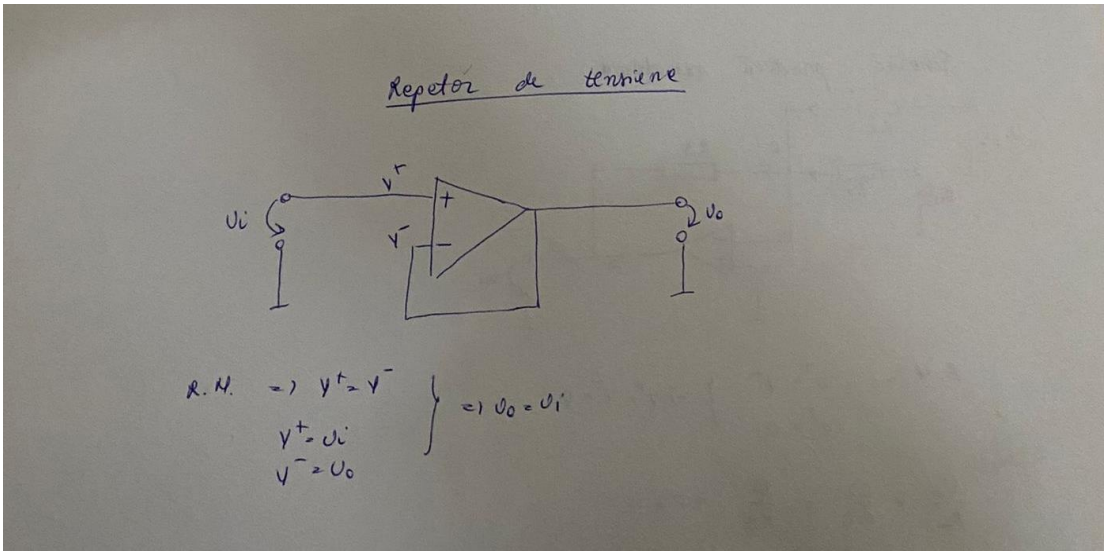
Sumator - circuitul 2:

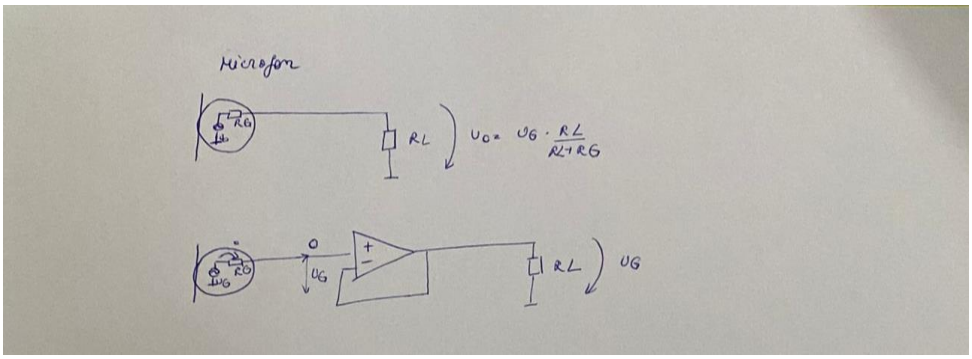
- Verificam daca avem reactie negativa. Ne imaginam o perturbatie la iesire si observam ca daca creste potentialul tensiunii de intrare, cel al tensiunii de iesire scade, asadar potentalele nodurilor de intrare sunt egale(V- = V+). Curentii se exprima in functie de tensiuni si rezistente.
- Dupa amplificatorul sumator legam unul inversor daca vrem ca U0 sa fie pozitiv.



Circuitul 3 :

- Avem reactie negativa -> cele 2 potentiale V- si V+ sunt egale.
- R4 mai mic -> Ui2 mai mare





Repetor de tensiune

Circuit 4 :

- Are R.N.-> V+ = V-
- Circuitul repeta tensiunea de intrare
- Microfon, se formeaza un divizor de tensiune
- Daca se adauga un repetor intre microfon si rezistenta. Intensitatea curentului de intrare in repetor este foarte mica. Tensiunea de iesire va fi tensiunea de generator, nu depinde de RL